ALP-ttbar Toy MC v1.1: Magnetic-Field Separation, Photon-Conversion Backgrounds and Vertex Resolution for Displaced Light ALP $\rightarrow e^+e^-$

(companion to arXiv:2610.38932)

Won Shik Paik (independent, personal project) · v1.1 · 31 October 2026

Companion to: J. Alimena, F. Blekman, C. Cazzaniga, X. Cid Vidal, J. Niedziela, L. Rygaard, R. Soto Rey, *Hunting for Long-Lived ALPs in Top Events via Displaced Dielectron Pairs at the LHC*, arXiv:2610.38932. Code: github.com/wspaik5/ALP_ttbar_Displaced_Dielectron_Toy_MC_arXiv2610.38932. This version (v1.1): doi:10.5281/zenodo.23833475. Previous version (v1.0): doi:10.5281/zenodo.23825276

Disclaimer. This is a simplified, numpy-only toy Monte Carlo written out of curiosity after reading the paper above. It is not affiliated with the authors or with the CMS Collaboration, uses parametrised ALP kinematics and assumed detector numbers, and should be read as qualitative trends only. The paper itself explicitly places detector-level reconstruction and instrumental backgrounds outside its scope; this note explores some of those effects in a rough way.

What is new in v1.1. v1.0 assumed a fixed radial vertex resolution (1 mm in the pixel region, 3 mm elsewhere) for the material veto against photon conversions. v1.1 replaces it with a per-pair resolution derived from the pair kinematics, optimises the veto width and tests non-Gaussian resolution tails (Sec. 3.4, Figure 4, Table 3). The conclusion for the lightest masses changes: the veto is much less robust than v1.0 suggested. All v1.0 figures and numbers are reproduced unchanged; Table 1 additionally lists 50 MeV.

1. Motivation

The paper extends the top-associated displaced-lepton strategy to $a \rightarrow e^+e^-$, reaching ALP masses down to 17 MeV, below the dimuon threshold. Two challenges flagged by the authors for the lowest masses are (i) the very small e^+e^- opening angle, evaluated in the paper with straight-line trajectories, and (ii) instrumental backgrounds such as interactions with detector material. The authors note that the opposite bending of e^+ and e^- in the CMS field may mitigate (i). This note quantifies (i) with and without the field, gives a rough look at (ii) via photon conversions, and, new in v1.1, asks how robust the proposed handle against (ii) is.

2. Toy model

ALP kinematics. $p_T(a) = 20\ \text{GeV} + \text{Exp}(80\ \text{GeV})$, $\eta \sim N(0, 1.6)$; exponential decay with lab length $\beta\gamma c\tau$; isotropic two-body decay boosted to the lab. The parametrisation was tuned so that the *shape* of the pre-selection efficiency over (m_a, $c\tau$) matches Table 3 of the paper (rms log-ratio 0.26); the absolute toy efficiency is a factor ~3.6 higher than the paper (unresolved; likely from effects not modelled, e.g. FSR and the full ttbar+a event).

Detector (barrel only, $|\eta| < 1.2$). 14 CMS Phase-1 barrel layer radii (4 pixel, 4 TIB, 6 TOB); uniform 3.8 T solenoid with exact helix–cylinder intersections; a pair is “resolved” if its two hits are separated by more than an assumed two-hit resolution (0.2 mm pixel, 0.4 mm strip) in at least 3 layers. Pre-selection follows the paper: 0.5 mm $< l_{xy} <$ 0.8 m.

Conversions. Isolated photons from ttbar+γ with an assumed $\sigma(p_T^{\gamma} > 20\ \text{GeV}) \approx 1$ pb, p_T^{-3} spectrum, 500 fb^{-1}; an assumed material map (beam pipe, pixel layers and supports, strip layers; $\approx 0.46\ X_0$ at $\eta = 0$, conversion probability $7/9 \cdot x/X_0$); Bethe–Heitler-like energy sharing. Track directions are smeared (0.3 mrad pixel-seeded / 0.8 mrad strip-only, plus 2 mrad·GeV/p multiple scattering, 2% momentum resolution). Vertex radii are smeared by a fixed 1 mm (pixel region) / 3 mm in v1.0 and by a per-pair resolution in v1.1 (Sec. 3.4). Cuts mirror the paper (min. electron $p_T > 1$ GeV) plus pair $p_T > 20$ GeV. Expected limits use a crude counting rule $N_{up} = \max(3, 2\sqrt{B})$, so only *ratios* are meaningful.

3. Results

3.1 Geometry and validation. Figure 1 shows the decay-position and boost effects behind the paper’s efficiency pattern, and the toy-vs-paper comparison of pre-selection efficiencies (after one global normalisation factor).

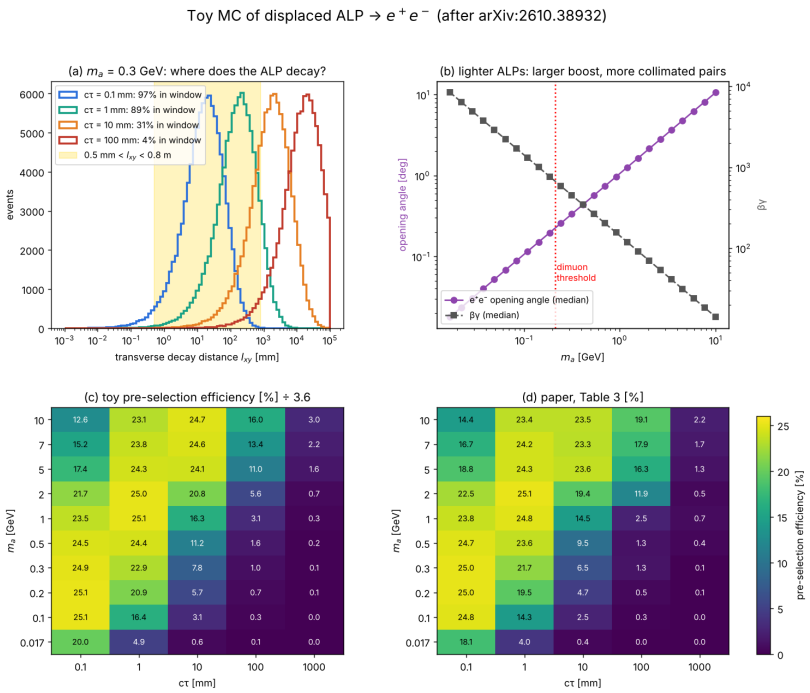


Figure 1. (a) Decay-distance distribution for m_a = 0.3 GeV; (b) median opening angle and boost vs. mass; (c) toy pre-selection efficiency divided by 3.6; (d) paper Table 3.

3.2 Magnetic-field separation. With straight lines, a 17 MeV pair stays below the assumed two-hit resolution in most layers; in the 3.8 T field the e^+ and e^- bend apart and reach centimetre separations by the outer tracker.

m_a	straight lines	3.8 T field
17 MeV	55%	97%
40 MeV	82%	97%
50 MeV	87%	97%
100 MeV	95%	97%
300 MeV	99.5%	98.6%

Table 1. Fraction of pre-selected pairs resolved in ≥ 3 barrel layers (cτ = 1 mm; results for cτ = 0.1 and 10 mm are within a few percent). This is a geometric criterion, not a tracking efficiency. Above ~0.2 GeV the field costs 1–2% because the softest electrons curl before reaching three layers.

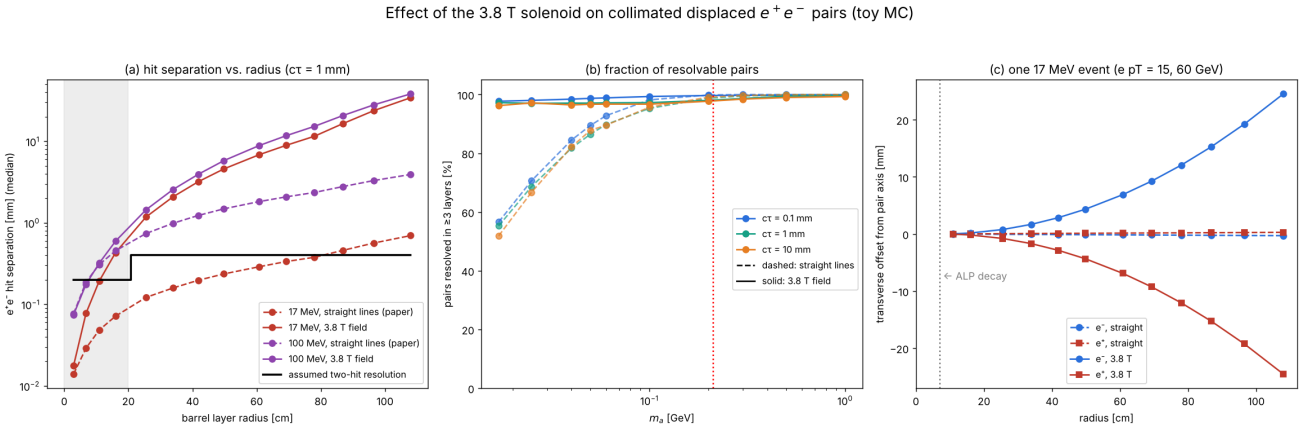


Figure 2. (a) Median hit separation per layer; (b) resolvable fraction vs. mass for three lifetimes; (c) one 17 MeV event.

3.3 Photon conversions (v1.0 settings). Under the assumptions above, about 9×10^4 conversions pass the paper-like cuts in the barrel, compared with tens of signal events at $\sigma\times B \sim 10^{-3}$ pb. Conversions cluster at material radii, whereas ALP decays are spread smoothly; the measured conversion mass is resolution-dominated and overlaps a 17 MeV signal, but not 100–200 MeV signals. In v1.0 the material veto removes a band of ±(half-width

+ $2\sigma_{fixed}$) around each material layer.

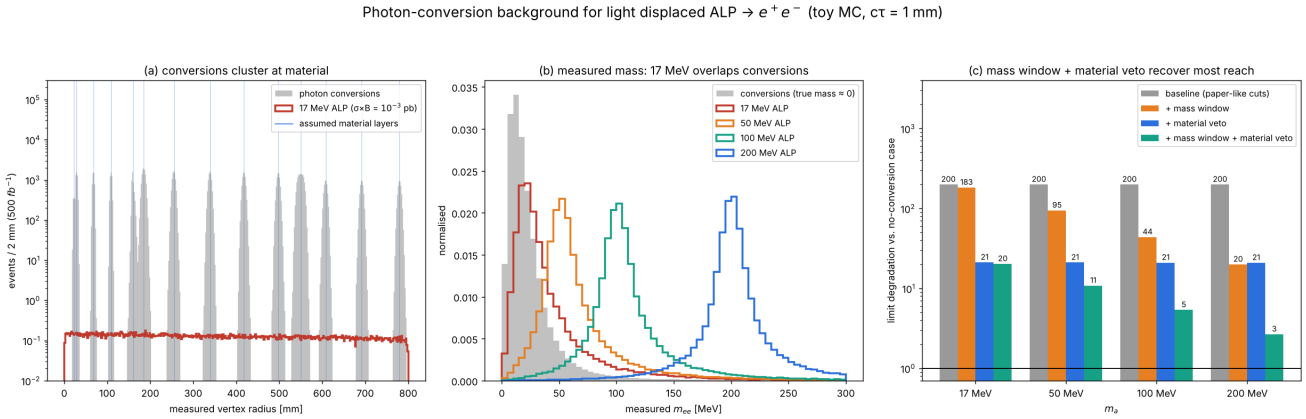


Figure 3. (a) Measured vertex radius; (b) measured m_{ee}; (c) degradation of the expected limit relative to the no-conversion case for successive selections (v1.0 settings, cτ = 1 mm).

m_a	baseline	+ mass window	+ material veto	both
17 MeV	200×	183×	21×	20×
50 MeV	200×	95×	21×	11×
100 MeV	200×	44×	21×	5×
200 MeV	200×	20×	21×	3×

Table 2. v1.0: factor by which the expected limit worsens because of conversions (fixed vertex resolution, 2σ veto band; toy, crude counting rule). Sec. 3.4 shows that these numbers depend strongly on the vertex-resolution assumptions.

3.4 New in v1.1: vertex resolution and robustness of the material veto. Two nearly parallel tracks only define a vertex if they open up. The toy now estimates a per-pair radial resolution from two constraints, combined as $\sigma_r^{-2} = \sigma_{free}^{-2} + \sigma_{tan}^{-2}$ (floor 0.05 mm):

- **crossing term**: $\sigma_{free} = \sqrt{2}\cdot[\sigma_{hit}^2 + (L_{gap}\sigma_\phi)^2]^{1/2} / \alpha_T$, where α_T is the transverse opening angle, L_{gap} the distance from the vertex to the first barrel layer, σ_ϕ the track direction resolution and σ_{hit} = 15 µm (pixel) / 35 µm (strip);
- **bending term**: $\sigma_{tan} = \sqrt{2}\cdot\sigma_\phi / k$ with $k = 1/R_1 + 1/R_2$, i.e. locating where the two oppositely bending tracks are parallel, as in a conversion fit.

Conversions ($\alpha_T \approx 0$) only receive the bending term. The veto band around each layer is widened by n_σ times the per-pair resolution, and n_σ (none, 0.5, …, 5) is optimised together with the mass window. As a robustness test, a “tails” variant gives 10% of pairs a true error three times larger than the resolution reported by the fit. The same optimisation is applied to the v1.0 fixed resolution for a like-for-like comparison.

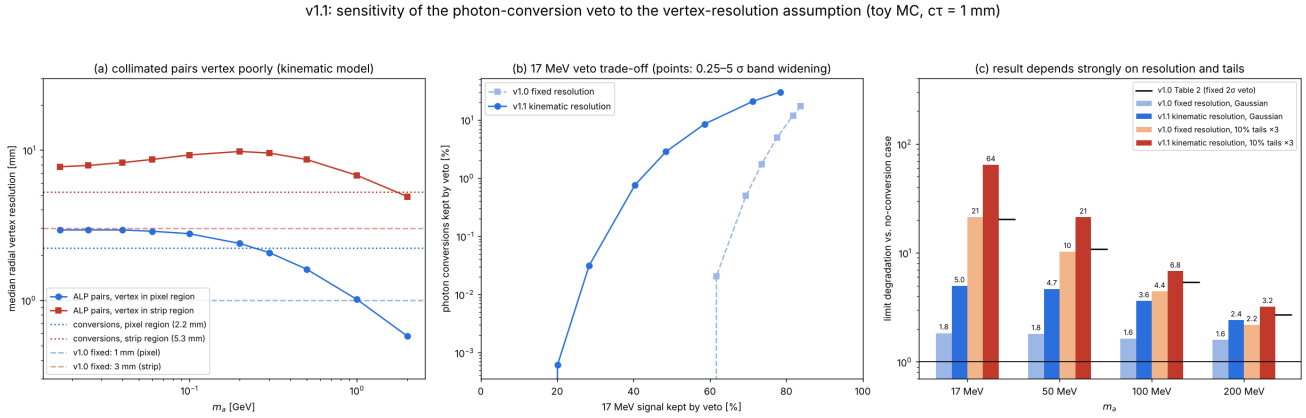


Figure 4 (new). (a) Median radial vertex resolution of ALP pairs vs. mass and of conversions, compared with the v1.0 fixed values; (b) 17 MeV signal kept vs. conversions kept as the veto band is widened from 0.25σ to 5σ (Gaussian errors); (c) limit degradation for four resolution scenarios with the veto width optimised; black markers: v1.0 Table 2.

m_a	v1.0 Table 2 (2σ band)	fixed, Gaussian	per-pair, Gaussian	fixed, tails	per-pair, tails
17 MeV	20×	1.8×	5.0×	21×	64×
50 MeV	11×	1.8×	4.7×	10×	21×
100 MeV	5.4×	1.6×	3.6×	4.4×	6.8×
200 MeV	2.7×	1.6×	2.4×	2.2×	3.2×

Table 3. Limit degradation from conversions with mass window and material veto, $c\tau$ = 1 mm. Except for the first column, the veto width is optimised. Values near 1.5–2× sit at the floor of the counting rule and rest on a few weighted conversion MC events.

Findings. (1) In this model, pairs below a few hundred MeV vertex about as poorly as conversions: ≈ 3 mm in the pixel region and ≈ 8–10 mm in the strip region (conversions: 2.2 and 5.3 mm), compared with the 1 / 3 mm assumed in v1.0; the v1.0 values correspond to pairs of roughly 1 GeV. (2) With the same optimised veto, the per-pair resolution worsens the 17–50 MeV result by a factor ≈ 2.5–3, mostly through lost signal (Fig. 4b). (3) The outcome is driven less by the core resolution than by the tails: with perfectly Gaussian errors a wide veto removes essentially all conversions, while 10% of 3× tails raise the 17 MeV degradation to ≈ 20–60×. The v1.0 value of 20× mainly reflected the fixed 2σ band. (4) Hence the v1.0 statement that a mass window plus material veto "recover most reach" holds only under optimistic resolution assumptions at 17–50 MeV; at 100–200 MeV the conclusion is stable (≈ 1.5–7× in all scenarios).

4. Takeaways

- The B field, neglected in the paper's straight-line treatment, substantially resolves the collimation issue at the lowest masses in this toy, consistent with the authors' remark.
- Photon conversions could be a dominant background in the sub-100 MeV region. At 17 MeV the mass does not discriminate, so a material-position veto, which itself relies on field-assisted vertexing of nearly parallel tracks, is the key handle.
- New in v1.1: for the lightest ALPs this veto is fragile. Its power depends on the vertex resolution of nearly collinear pairs and, even more, on the size of non-Gaussian resolution tails; in this toy the 17 MeV loss in reach ranges from ≈ 2× to ≈ 60×.
- A realistic estimate needs the material map, the resolution and tails of conversion-like vertex fits for high-p_T collinear pairs, hit-pattern information, and the reconstruction efficiency for displaced, collimated tracks, all of which require detector-level simulation.

5. Limitations

Parametrised (not generated) ALP and photon kinematics; barrel only; no FSR, pile-up, track-finding efficiency, calorimeter or isolation modelling; assumed material budget and resolutions; ttbar+γ rate is an order-of-magnitude assumption; other backgrounds from the paper (e.g. heavy-flavour decays) are not included; the factor ~3.6 normalisation offset relative to the paper is not understood. The v1.1 vertex-resolution model is an error-propagation estimate rather than a vertex fit: it ignores hit-pattern constraints and two-fold intersection ambiguities and treats the pair direction as radial; the tail fraction and scale are illustrative, and the veto width is optimised on the same samples used to evaluate it. Absolute numbers may be off by large factors; trends are the intended content.

6. Reproducibility

All figures and numbers are reproduced by `./run_all.sh` (Python 3, numpy, matplotlib; about 1 minute). New in v1.1: `vtx.py` (resolution model) and `plot_vertex.py` (Figure 4); `bkg.py` keeps the v1.0 behaviour by default. Repository: github.com/wspaik5/ALP_ttbar_Displaced_Dielectron_Toy_MC_arXiv2610.38932. Fixed random seeds are used throughout.